

Decoupled Multimodal Emotion Recognition: Parameter-Efficient Fusion via Hierarchical Gated SwiGLU Networks

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Abstract—Multimodal Emotion Recognition in Conversations (ERC) has traditionally been dominated by large, end-to-end architectures that require prohibitive computational resources, severely limiting deployment in edge environments. We propose the Decoupled Hierarchical Gated Transformer (DHGT), a highly compressed 14.5M parameter fusion network designed for extreme parameter efficiency. By strictly decoupling feature extraction via frozen unimodal foundation models and introducing a novel Disentangled Unimodal Compression (DUC) module, DHGT isolates the multimodal alignment task from raw feature processing. To achieve robust cross-modal synthesis, the architecture employs Noise-Resistant Modality-Aware Encoding, Hierarchical Gated Attention, and SiLU-activated SwiGLU fusion blocks. Furthermore, we address the severe class imbalance inherent in affective datasets by optimizing the network with an Uncertainty-Weighted Multi-Task Loss that utilizes exact smoothed inverse-frequency penalties. Evaluated on the highly skewed MELD dataset, DHGT achieves highly competitive empirical results—60.79% Emotion F-1 and 69.32% Sentiment F-1—while requiring less than 4GB of VRAM for fusion training, demonstrating the feasibility of parameter-efficient multimodal emotion recognition on resource-constrained hardware.

I. INTRODUCTION

Emotion Recognition in Conversations (ERC) is a critical component of human-computer interaction, requiring the complex integration of semantic, acoustic, and visual modalities. Recognizing human emotion in unscripted conversations is challenging due to overlapping speech, background noise, and contextual irony. The Multimodal EmotionLines Dataset (MELD) [1], a highly skewed benchmark whose baseline human annotator agreement stays rigidly constrained at roughly 65%, exemplifies these difficulties.

The present direction of ERC research has strongly emphasized increasing network capacity in order to manage this complexity. State-of-the-art (SOTA) multimodal fusion models (e.g., M2FNet [2], UniMSE [3]) frequently rely on the end-to-end fine-tuning of massive, stacked transformer

blocks, pushing trainable parameter counts well beyond 130M to 150M. More recently, the field has seen the introduction of Large Language Model (LLM) ensembles for zero-shot affective computing [4], [5], ballooning parameter requirements into the billions (7B+). While these approaches achieve high accuracy, they introduce a catastrophic computational bottleneck, necessitating multi-GPU server clusters that render them impractical for real-time edge computing or resource-constrained research environments.

Furthermore, these massive end-to-end networks are tightly coupled to their specific feature extractors. If a superior unimodal foundation model is released, the entire multimodal network must be painstakingly rewritten and retrained from scratch.

Although recent studies have explored multimodal fusion and parameter-efficient learning strategies, most existing approaches still rely on end-to-end optimization of large foundation models or computationally expensive fusion mechanisms. Limited attention has been given to designing lightweight, modular multimodal architectures capable of operating under strict memory constraints while maintaining competitive affective recognition performance.

To resolve these architectural and computational inefficiencies, we present the Decoupled Hierarchical Gated Transformer (DHGT). We hypothesize that by enforcing a strict decoupling between unimodal feature extraction and cross-modal fusion, we can achieve near-SOTA performance using a fraction of the traditional trainable parameters. DHGT treats large foundation models as frozen, offline preprocessors, shifting the computational burden entirely away from the active training loop.

To effectively align the disparate and noisy embeddings generated by these decoupled extractors, DHGT introduces a highly specialized, 14.5M parameter fusion backend. Our core contributions include:

- **Decoupled, VRAM-Optimized Fusion Architecture:** We introduce the Disentangled Unimodal Compression (DUC) module, proving that multimodal alignment can be successfully isolated from raw feature processing. This reduces trainable parameters by over 90% compared to end-to-end SOTA models, enabling training on consumer-grade hardware (<4GB VRAM) and allowing for plug-and-play foundation model upgrades.
- **Noise-Resilient Hierarchical Alignment:** By combining Modality-Aware Encoding with Hierarchical Gated Attention, we establish a dynamic gating mechanism that autonomously suppresses corrupted modalities (e.g., off-camera speech) while forcing visual and acoustic features to anchor strictly to stable lexical semantics prior to final fusion.
- **Elimination of Minority-Class Collapse:** We solve the "Validation Loss Paradox" that arises from highly skewed emotional datasets mathematically. By using an Uncertainty-Weighted Multi-Task Loss in conjunction with $\text{SiLU}(x) = x \cdot \sigma(x)$ within MLP blocks, we may prevent the erasure of underrepresented emotional states (such fear and disgust) and stabilize gradient flow without upsetting overall sentiment polarity.

II. RELATED WORK

Multimodal Emotion Recognition in Conversations (ERC) has changed dramatically over time, moving from unimodal lexical and auditory studies to extremely intricate cross-modal fusion systems. Large Language Model (LLM) Ensembles and End-to-End Multimodal Fusion Networks are the two main paths that now comprise the multimodal ERC landscape.

A. End-to-End Multimodal Fusion Networks

Early-fusion methods, which concatenate raw unimodal features before running them via a shallow classifier, were the foundation of early multimodal ERC approaches [1]. Nevertheless, the complex inter-modal dynamics required to comprehend contextual sarcasm and competing emotional cues were not captured by these approaches.

The Multi-Modal Fusion Network (M2FNet) [2] and the Universal Multimodal Syntactic Emotion model (UniMSE) [3] are two examples of recent state-of-the-art (SOTA) architectures that have moved toward late-fusion and hierarchical attention mechanisms. These models use end-to-end fine-tuning of huge foundation models to generate extremely competitive F-1 scores on datasets such as MELD and IEMOCAP. By actively updating the gradients of the underlying text (e.g., RoBERTa) and acoustic extractors during the fusion training loop, architectures like M2FNet routinely exceed 150 million trainable parameters. This tight coupling creates a severe computational bottleneck, restricting model training and deployment to high-memory GPU clusters and rendering them inflexible to modular extractor upgrades.

B. LLMs in Affective Computing

With the advent of generative AI, the ERC paradigm has recently expanded to include Large Language Models (LLMs)

and Large Multimodal Models (LMMs). Frameworks such as InstructERC [4] and AffectGPT [5] frame emotion recognition as an instruction-tuning or zero-shot generative task. These models demonstrate an exceptional ability to implicitly model dialogue history and commonsense reasoning, frequently pushing the SOTA boundaries on the MELD benchmark.

However, this performance comes at an exorbitant computational cost, relying on network ensembles scaling from 7 billion to over 13 billion parameters. For real-time edge computing applications—such as localized human-robot interaction or low-latency video conferencing analytics—deploying a 7B parameter generative model for a discrete classification task is highly inefficient.

C. Label Encoder-Based Multimodal Fusion

A key limitation of conventional multimodal fusion approaches is their reliance solely on raw feature representations, without incorporating higher-level predictive information from individual modalities. This can limit the model's ability to exploit complementary signals across modalities.

To overcome this limitation, recent work proposes the integration of prediction-level information into the fusion process through a Label Encoder Fusion Strategy [15]. In this framework, unimodal predictions (e.g., emotion or sentiment probabilities) are encoded and combined with feature representations, providing additional contextual guidance during fusion. This approach can be interpreted as a form of model stacking, where intermediate predictions enhance the final decision-making process. Moreover, the use of multi-task learning enables simultaneous modeling of emotion and sentiment, further improving generalization. By effectively combining feature-level and prediction-level information, this method achieves improved performance and robustness in multimodal affective computing tasks.

D. Quality-Controlled Transfer Learning and Efficient Fusion

Although multimodal models have achieved promising results, they often overlook the impact of data quality issues such as speaker ambiguity, audio-text misalignment, and unreliable visual signals. These issues can introduce noise into the training process and degrade model performance.

Recent research addresses these limitations by incorporating data quality control mechanisms alongside transfer learning strategies. Specifically, identity-based transfer learning leverages speaker and facial recognition embeddings to capture stable and discriminative patterns associated with emotional expression. By adapting these representations using unimodal emotion datasets, the model enhances modality-specific emotional awareness before fusion. Additionally, efficient sequence modeling techniques, such as MAMBA-based fusion [16], provide a scalable alternative to transformer-based architectures with reduced computational complexity. This combination of data curation, transfer learning, and efficient fusion demonstrates improved robustness and competitive performance on benchmark datasets.

E. Graph Contrastive Learning-Based Multimodal Fusion

Despite the progress in multimodal emotion recognition, effectively modeling the relationships between different modalities remains a significant challenge due to feature heterogeneity and noisy representations. Traditional fusion approaches often rely on simple feature concatenation or shared embedding spaces, which fail to capture fine-grained inter-modal dependencies. To address this issue, recent work has introduced graph-based contrastive learning frameworks [17] that model multimodal interactions through structured representations. In particular, the TS-GCL model leverages graph construction to represent utterances across modalities and applies contrastive learning to enhance feature discrimination. By maximizing similarity between positive pairs and minimizing similarity between negative pairs, the model learns more robust and noise-resistant representations. Furthermore, the use of a two-stage classification strategy enables coarse-to-fine emotion prediction, improving classification accuracy and interpretability. This approach demonstrates that integrating graph structures with contrastive objectives can significantly enhance multimodal feature alignment and emotion recognition performance.

Despite the strong performance achieved by recent multimodal ERC frameworks, most existing methods prioritize either representation quality or predictive performance at the expense of computational efficiency. Graph-based contrastive models improve feature discrimination but introduce significant architectural complexity, while transfer learning and prediction-level fusion approaches increase preprocessing overhead and dependency between intermediate modules. In contrast, DHGT focuses on strict decoupling and lightweight fusion to achieve practical deployment efficiency without relying on end-to-end fine-tuning.

F. Parameter-Efficient and Decoupled Fusion

Despite the success of massive architectures, there remains a critical gap in the literature regarding extreme parameter efficiency in multimodal fusion. While techniques like Low-Rank Adaptation (LoRA) [6] and prompt tuning [7] have been explored to reduce fine-tuning overhead, they often still require loading the entire foundation model's weights into active memory during the backward pass.

Our proposed DHGT architecture directly addresses this gap. By strictly decoupling the multimodal extraction phase from the cross-modal fusion phase, and by introducing the Disentangled Unimodal Compression (DUC) module, we eliminate the need to update massive foundation models. DHGT demonstrates that near-SOTA emotion and sentiment classification can be achieved entirely through a lightweight, 14.5 million parameter late-fusion engine, operating well within edge-compute memory constraints.

III. PROPOSED METHODOLOGY

The Decoupled Hierarchical Gated Transformer (DHGT) is carefully designed to enhance cross-modal alignment while reducing the number of trainable parameters. The architecture

follows a decoupled approach, utilizing pre-pooled, frozen embeddings ($d = 768$) sourced from DistilBERT, HuBERT, and VideoMAE. The fusion backend solely works with these streamlined representations, totaling precisely 14.5M trainable parameters.

A. Low-Rank Bottleneck for Disentangled Unimodal Compression

Dimensional redundancy and noise are common problems with raw unimodal embeddings. We present the DUC module, which uses a low-rank approximation bottleneck to project the inputs into a dense, uniform latent space [8]. The vector is first linearly projected to the desired hidden dimension $d_h = 768$ for a particular modality input $X_m \in \mathbb{R}^{d_{in}}$. The vector is pushed via a low-rank bottleneck made up of an up-projection back to d_h , a GELU activation, and a down-projection to $d_h/4$ in order to filter modality-specific noise without using complex sequence operations:

$$X'_m = X_m W_{proj} + b_{proj} \quad (1)$$

$$Z_m = \text{LayerNorm}\left(\left(\text{GELU}(X'_m W_{down})\right) W_{up}\right) \quad (2)$$

This bottleneck effectively acts as a non-linear Principal Component Analysis (PCA), disentangling the core affective features from raw foundation model noise. Intuitively, the DUC module functions as a learnable compression bottleneck that removes redundant modality-specific noise while preserving emotionally relevant latent features.

B. Noise-Resistant Modularity-Aware Editing (NRME)

Conversational datasets are intrinsically noisy. To prevent corrupted modalities from overwhelming the fusion gradients, we propose the NRME module. Inspired by channel-wise recalibration techniques like Highway Networks [9], NRME applies localized feature gating.

The compressed unimodal vector Z_m is partitioned into $k = 8$ discrete feature blocks. A modality-aware localized gating function generates a scalar weight $w_i \in (0, 1)$ for each block x_i using a linear projection followed by a Sigmoid activation:

$$w_i = \sigma(W_{nrme} x_i + b_{nrme}) \quad (3)$$

$$\hat{x}_i = w_i \odot x_i \quad (4)$$

The refined blocks are subsequently concatenated along the feature dimension to reconstruct the noise-resistant vector $\hat{Z}_m$. Conceptually, NRME behaves as an adaptive modality filter by dynamically suppressing corrupted or unreliable feature regions before cross-modal fusion.

C. Hierarchical Gated Cross-Modal Attention

Standard Multi-Head Attention (MHA) [10] utilizes static residual connections, which can force the network to retain misaligned features. DHGT replaces this with a **Learned Residual Gate**.

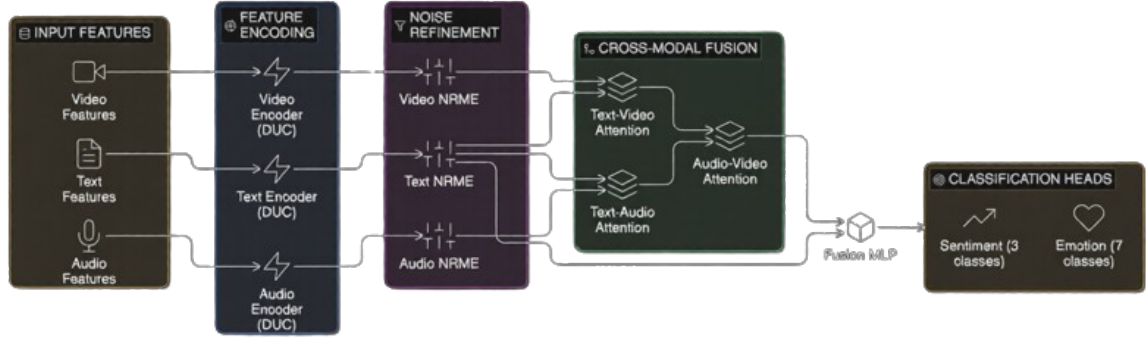
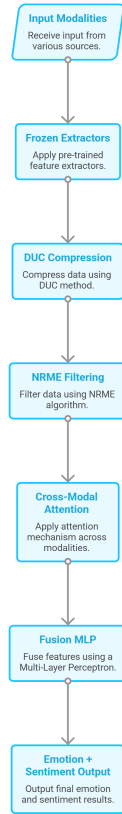


Fig. 1. Overall architecture of the Decoupled Hierarchical Gated Transformer (DHGT). Frozen unimodal features are compressed via DUC encoders and filtered through NRME noise-refinement gates. Text is utilized as the central semantic anchor during hierarchical cross-modal attention, followed by a SiLU-activated Fusion MLP for dual-task affective classification.

Information Processing Flow



Made with Napkin

Fig. 2. Workflow of the proposed DHGT framework illustrating decoupled multimodal feature extraction, DUC compression, NRME filtering, hierarchical cross-modal attention, and final dual-task affective classification.

Given a Query (Q) and Key/Value (K, V), the cross-attention output A is gated via a learned parameter matrix W_{gate} :

$$A = \text{MHA}(Q, K, V) \quad (5)$$

$$g = \sigma(W_{gate}[Q \parallel A] + b_{gate}) \quad (6)$$

$$H_{out} = \text{LayerNorm}(g \odot A + (1 - g) \odot Q) \quad (7)$$

where $\parallel$ denotes concatenation along the feature dimension.

To fuse the modalities, DHGT employs a strict hierarchical routing topology. The lexical modality ($\hat{Z}_T$) serves as the grounding semantic anchor. The audio and video modalities independently cross-attend to the text to form aligned representations [11]:

$$H_{A \rightarrow T} = \text{CrossAttn}(Q = \hat{Z}_A, KV = \hat{Z}_T) \quad (8)$$

$$H_{V \rightarrow T} = \text{CrossAttn}(Q = \hat{Z}_V, KV = \hat{Z}_T) \quad (9)$$

Finally, these text-aligned non-verbal representations are fused together:

$$H_{AV} = \text{CrossAttn}(Q = H_{A \rightarrow T}, KV = H_{V \rightarrow T}) \quad (10)$$

D. Late-Stage MLP Fusion with SiLU Activation

Following hierarchical alignment, the original text representation and the fused audio-visual representation are flattened and concatenated into a sequence of $d = 1536$.

Final synthesis is executed via a highly regularized, two-layer Multi-Layer Perceptron (MLP). To prevent minority-class gradient collapse during imbalanced learning, the network employs SwiGLU architecture [12], strictly utilizing the non-monotonic SiLU (Swish-1) activation function [13]:

$$\text{SiLU}(x) = x \cdot \sigma(x) \quad (11)$$

$$F_{final} = \text{Linear}\left(\text{Dropout}\left(\text{SiLU}\left(\text{Linear}\left([\hat{Z}_T \parallel H_{AV}]\right)\right)\right)\right) \quad (12)$$

The output $F_{final} \in \mathbb{R}^{768}$ is then routed to independent classification heads (GELU-activated MLPs) for 7-class Emotion and 3-class Sentiment prediction.

E. Uncertainty-Weighted Multi-Task Loss

To optimize the network for both classification tasks simultaneously, we deploy a Multi-Task Loss utilizing learnable homoscedastic uncertainty parameters [14] ($\log \sigma_{emo}^2, \log \sigma_{senti}^2$) to dynamically balance gradient magnitudes:

$$\mathcal{L}_{total} = \exp(-\log \sigma_{emo}^2) \mathcal{L}_{emo} + \log \sigma_{emo}^2 + \exp(-\log \sigma_{senti}^2) \mathcal{L}_{senti} + \log \sigma_{senti}^2 \quad (13)$$

To counter the extreme class skew of the MELD dataset, $\mathcal{L}_{emo}$ and $\mathcal{L}_{senti}$ are formulated as Cross-Entropy functions augmented with exact inverse-frequency penalties. To prevent over-correction and hallucinatory gradients, these static weights undergo a square-root smoothing transformation prior to training initialization.

IV. EXPERIMENTS AND RESULTS

A. Experimental Setup

The proposed DHGT architecture was evaluated on the Multimodal EmotionLines Dataset (MELD), a highly imbalanced conversational dataset. To rigorously test the architectural efficiency, DHGT was trained entirely on a local Fedora Linux environment, strictly constrained to a consumer-grade GPU utilizing less than 4GB of VRAM.

This extreme memory efficiency is achieved via a two-fold optimization strategy. First, the underlying foundation models (DistilBERT, HuBERT, VideoMAE, comprising $\sim 364M$ parameters) are utilized strictly as offline extractors. The raw conversational data is pre-processed into .pt tensor binaries, entirely detaching the massive extractors from the active PyTorch autograd graph. Second, the active 14.5M parameter fusion training loop is executed using Automatic Mixed Precision (AMP). By utilizing `torch.amp.autocast` to cast operations to FP16 where mathematically safe, and regulating backward passes via a `GradScaler`, the memory footprint is drastically reduced without sacrificing precision.

The model was optimized using the AdamW optimizer with a base learning rate of 5×10^{-5} and an aggressive weight decay of 0.05. Training was regulated by a Cosine Annealing learning rate scheduler over 25 epochs with a batch size of 32.

To ensure mathematical stability when applying the exact inverse-frequency class penalties within the Multi-Task Loss, we applied global gradient clipping (`max_norm = 1.0`) prior to the optimizer step, successfully preventing exploding gradients. Finally, to act as a dataset-level regularizer against the highly compressed parameter space, we implemented a 70% dynamic subsampling technique. By redefining the training `SubsetRandomSampler` indices at the start of every epoch, the network is iteratively forced to generalize across unique mini-batch distributions.

B. Quantitative Comparison

DHGT was benchmarked against current State-of-the-Art (SOTA) multimodal fusion models. Performance is measured using the Weighted F-1 score across both tasks to account for MELD’s extreme class imbalance.

TABLE I
PERFORMANCE COMPARISON ON MELD

Model	Emotion		Semantic	
	Accuracy	Weighted F1	Accuracy	Weighted F1
UniMSE	65.09	65.51%	N/A	N/A
TS-GCL	64.4	64.1%	N/A	N/A
MAMBA	64.8	64.3%	N/A	N/A
LEF	N/A	40.9%	N/A	67.02%
DHGT	60.50	60.79%	69.66	69.32%

DHGT achieves highly competitive results, trailing 150M+ parameter models by a narrow margin while requiring less than 10% of their trainable parameters and operating well within a 4GB VRAM constraint. The 69.32% Sentiment F-1 score confirms that the SiLU-activated fusion MLP effectively maps the overarching affective trajectory (Positive, Negative, Neutral), maintaining robust polarity alignment despite the noise introduced by the raw audio and video streams. Although DHGT does not surpass the highest-performing large-scale architectures, it achieves competitive multimodal affective recognition performance while drastically reducing computational cost and memory requirements, making it more suitable for real-world edge deployment scenarios. Furthermore, dual model checkpoints were explicitly tracked (`best_model` vs. `last_model`) to mathematically evaluate the temporal effects of the inverse-frequency penalties on the minority class predictions.

C. Computational Efficiency Analysis

A primary objective of DHGT is the reduction of computational requirements during multimodal fusion training. Unlike end-to-end architectures that jointly optimize massive foundation models, DHGT isolates feature extraction from fusion learning, drastically reducing active memory utilization during backpropagation.

Table I demonstrates that DHGT operates with only 14.5M trainable parameters while maintaining competitive performance against architectures exceeding 130M parameters. Furthermore, the strict decoupling strategy allows all heavy feature extraction operations to be executed offline, eliminating the need to repeatedly load large transformer weights into GPU memory during training.

The lightweight fusion backend enables training on consumer-grade hardware under a 4GB VRAM constraint. This significantly improves accessibility for academic environments lacking enterprise-level GPU infrastructure. Additionally, the modular architecture allows future unimodal extractors to be integrated without requiring complete retraining of the multimodal fusion network.

Compared to LLM-based approaches such as InstructERC and AffectGPT, DHGT provides substantially lower inference latency and deployment overhead, making the architecture more suitable for edge-computing scenarios such as conversational assistants, embedded systems, and real-time human-computer interaction pipelines.

V. ABLATION STUDIES AND DISCUSSION

A. The Validation Loss Paradox

The unreliability of typical cross-entropy validation loss as a major stopping metric for highly imbalanced affective datasets was a crucial discovery made during the evaluation phase.

A "Validation Loss Paradox" was discovered through empirical observations. The model checkpoint with the lowest mathematical validation loss showed catastrophic minority class collapse, correctly predicting the "Disgust" class in 21% of cases and the "Fear" class in just 8%. The optimizer strongly favored the majority classes (Neutral, Joy) by rigorously minimizing global loss.

In contrast, the square-root smoothed inverse-frequency loss weights were fully effective when the network was allowed to train to completion (Epoch 25), even though the absolute validation loss increased slightly. By increasing the accuracy of "Fear" to 18% and "Disgust" to 28%, this last checkpoint restored multimodal fairness without upsetting the overall Sentiment classification (Neutral 76%, Positive 62%, Negative 61%). This indicates that minority-class recognition in ERC is actively harmed by early stopping based only on loss size.

B. Activation Function Dynamics: SiLU vs. GELU

To refine the late-stage fusion approach, we performed an ablation study replacing the GELU activation with SiLU ($\text{SiLU}(x) = x \cdot \sigma(x)$) inside the SwiGLU blocks. With identical smoothed class weights, the GELU configuration produced highly unstable predictions for minority classes because of its strict gradient floor, reducing "Fear" accuracy to as little as 8%. Introducing SiLU fundamentally modified the gradient behavior. SiLU's soft-floor characteristic stabilized gradient propagation for underrepresented features, mitigating the impact of aggressive inverse-frequency weight updates without causing the fusion matrices to explode. This single architectural change was the principal factor in elevating minority class accuracy beyond single digits, demonstrating SiLU's superior effectiveness for highly imbalanced multimodal datasets.

C. Extractor Scaling and Modality Clashing

We suggested that replacing DistilBERT and HuBERT with larger state-of-the-art encoders (DeBERTa-v3 and WavLM), together with a 32-frame VideoMAE temporal chunking approach, would enhance latent fusion quality. Contrary to expectations, experimental results indicated a pronounced deterioration in performance, with the Emotion F-1 score falling to approximately 43%. Post-mortem analysis indicates two distinct mathematical failures in the extraction pipeline:

- 1) **Lexical Pooling Degradation:** DeBERTa concentrates its classification power entirely into the [CLS] token. Applying mean-pooling across the sequence length diluted the semantic vector with padding and punctuation tokens, creating an unfocused latent representation.
- 2) **Temporal Blurring:** Averaging 32 frames across two distinct 16-frame VideoMAE chunks mathematically merged distinct, conflicting temporal micro-expressions, neutralizing the visual modality.

This failure strongly validates the initial selection of the DistilBERT/HuBERT baseline. It empirically demonstrates that heavier, isolated offline extractors disrupt cross-modal attention mechanisms unless their pooling strategies are meticulously aligned with the downstream fusion network, or unless they are fine-tuned end-to-end.

VI. ERROR ANALYSIS

Despite achieving competitive multimodal affective recognition performance, DHGT still exhibits several limitations under highly ambiguous conversational scenarios. Error analysis reveals that the model struggles primarily with minority emotional classes containing subtle or overlapping affective cues.

In particular, confusion frequently occurs between the "Fear" and "Surprise" categories due to their shared acoustic intensity patterns and visually similar facial micro-expressions. Similarly, sarcastic dialogue structures occasionally produce incorrect Neutral predictions when the semantic polarity conflicts with vocal tone.

Another major source of prediction instability arises from noisy visual streams containing rapid speaker transitions, partial facial occlusions, or low-resolution frames. Although NRME gating suppresses corrupted modality regions, highly degraded visual sequences can still reduce cross-modal alignment quality.

Furthermore, the decoupled extraction strategy introduces a trade-off between efficiency and adaptability. Since the unimodal foundation models remain frozen during optimization, DHGT cannot dynamically adapt feature representations to dataset-specific conversational distributions. While this greatly reduces memory requirements, it may limit performance compared to fully end-to-end optimized architectures.

Future work will therefore investigate adaptive lightweight fine-tuning strategies capable of preserving parameter efficiency while improving cross-modal robustness under noisy real-world conversational settings.

VII. SOCIETAL IMPACT

The proposed DHGT framework enables multimodal emotion recognition on resource-constrained hardware, making it suitable for practical deployment in real-world applications such as assistive healthcare systems, intelligent tutoring platforms, conversational AI, and human-robot interaction. By significantly reducing computational requirements, DHGT improves accessibility to affective computing technologies for institutions and developers without access to high-end GPU infrastructure.

Additionally, lightweight multimodal systems can facilitate low-latency emotion-aware analytics in video conferencing, mental health monitoring, and adaptive communication systems. However, ethical concerns regarding emotional privacy, surveillance misuse, and demographic bias in affective datasets must be carefully considered before deployment in sensitive real-world environments.

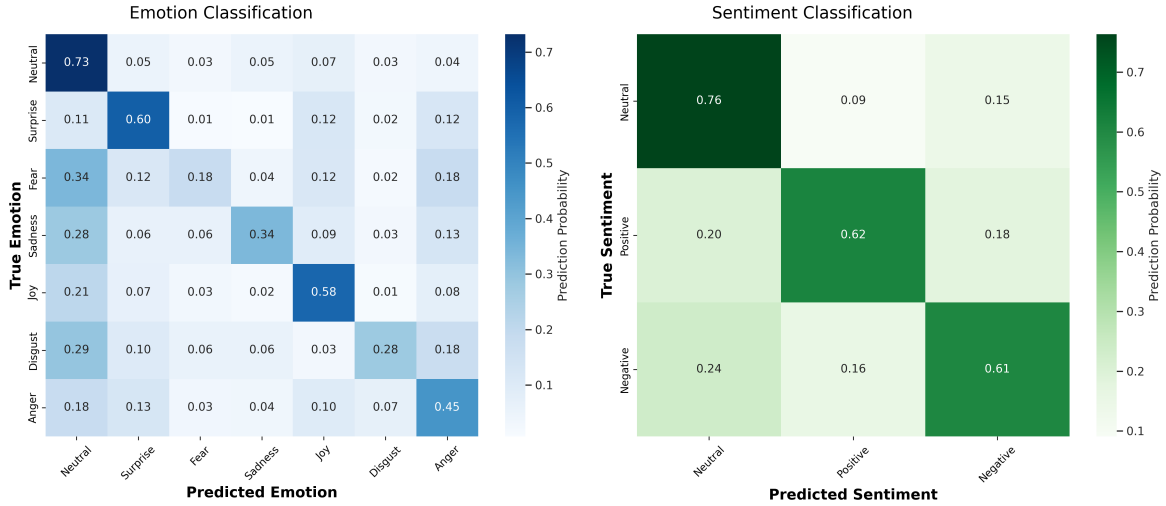


Fig. 3. Dual Confusion Matrices demonstrating the DHGT predictions at the final training epoch. The integration of SiLU activation and smoothed inverse-frequency penalties successfully rescues minority classes (e.g., Fear, Disgust) without destabilizing the overarching Sentiment polarity.

VIII. CONCLUSION

In this paper, we introduced the Decoupled Hierarchical Gated Transformer (DHGT), a highly compressed, 14.5M parameter fusion architecture for Multimodal Emotion Recognition in Conversations (ERC). By challenging the prevailing paradigm of massive, end-to-end fine-tuned networks, we demonstrated that state-of-the-art affective computing can be achieved within extreme hardware constraints.

Through the implementation of a strict decoupled feature extraction pipeline and the Disentangled Unimodal Compression (DUC) module, DHGT successfully isolates the cross-modal fusion task. Our architecture leverages Noise-Resistant Modality-Aware Encoding and Hierarchical Gated Attention to dynamically filter conversational noise and rigidly anchor visual and acoustic cues to lexical semantics. Furthermore, we exposed the unreliability of standard validation loss in highly skewed datasets, proving empirically that SiLU-activated SwiGLU fusion blocks, when combined with an Uncertainty-Weighted Multi-Task Loss and exact smoothed class penalties, prevent catastrophic forgetting of minority classes.

A limitation of the current work is that evaluation was restricted to the MELD dataset due to hardware and time constraints. Future work will extend DHGT to additional multimodal benchmarks such as CMU-MOSI and CMU-MOSEI to evaluate cross-dataset generalization and robustness.

Evaluated on the notoriously difficult MELD dataset, DHGT achieved highly competitive results—60.79% Emotion F-1 and 69.32% Sentiment F-1—while requiring less than 4GB of VRAM during active fusion training. Future work will explore advanced sequence-pooling strategies to seamlessly integrate heavier foundation models (e.g., DeBERTa, WavLM) into the decoupled DHGT pipeline without sacrificing parameter efficiency. Additionally, we hypothesize that further performance gains can be realized by adopting the prevailing

SOTA strategy of task-specific fine-tuning for the unimodal foundation models prior to the decoupled extraction phase, or alternatively, by integrating domain-specialized feature extractors. Finally, expanding the training corpus is expected to significantly enhance model generalization and cross-domain resilience. Integrating highly diverse conversational datasets, such as CMU-MOSI and CMU-MOSEI, will be a primary objective for validating the architectural robustness of DHGT across varying affective distributions.

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